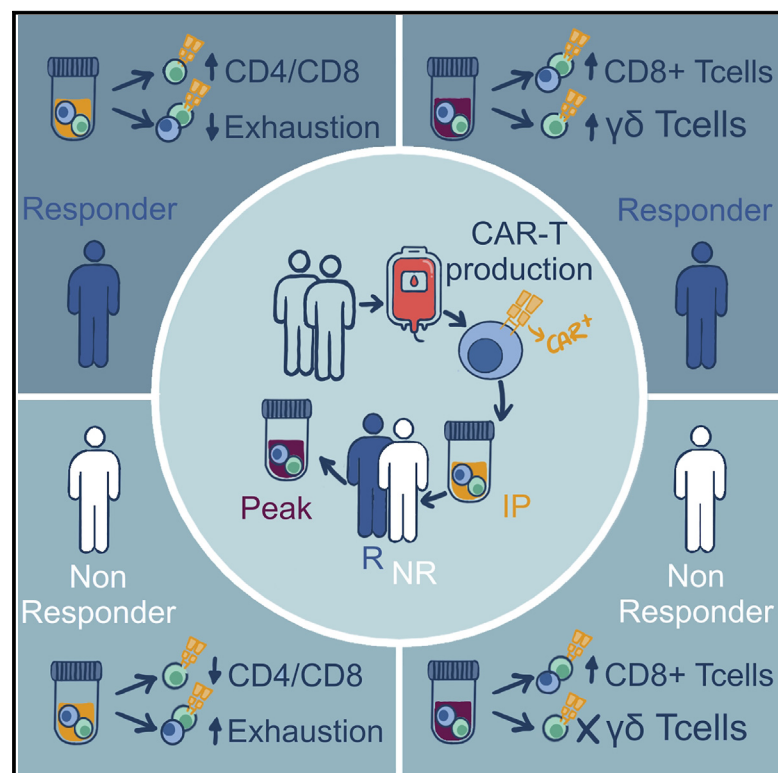


Integrative single-cell multi-omics of CD19-CAR^{pos} and CAR^{neg} T cells suggest drivers of immunotherapy response in B cell neoplasias

Graphical abstract



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In brief

Guerrero-Murillo and Rill-Hinarejos et al. demonstrate that phenotypic, clonal, and functional heterogeneity of CAR-T cells affect outcomes in B-ALL patients treated with CD19CAR-T therapy. Differences in CAR^{pos} and CAR^{neg} T cell dynamics, including CD4 ratios, cytotoxic CD8⁺, and γδT cell expansion, are associated with treatment efficacy across multiple patient cohorts.

Highlights

- CD4/CD8 CAR^{pos} T cell ratio predicts therapy outcome and CAR-T cell persistence
- Pre-existing T cell exhaustion in input cells may reduce CAR-T treatment efficacy
- CAR^{pos} and CAR^{neg} synergize to drive TCR diversification and CD8 cytotoxic expansion
- Expansion of cytotoxic γδT cells strongly correlates with treatment success



Article

Integrative single-cell multi-omics of CD19-CAR^{pos} and CAR^{neg} T cells suggest drivers of immunotherapy response in B cell neoplasias

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SUMMARY

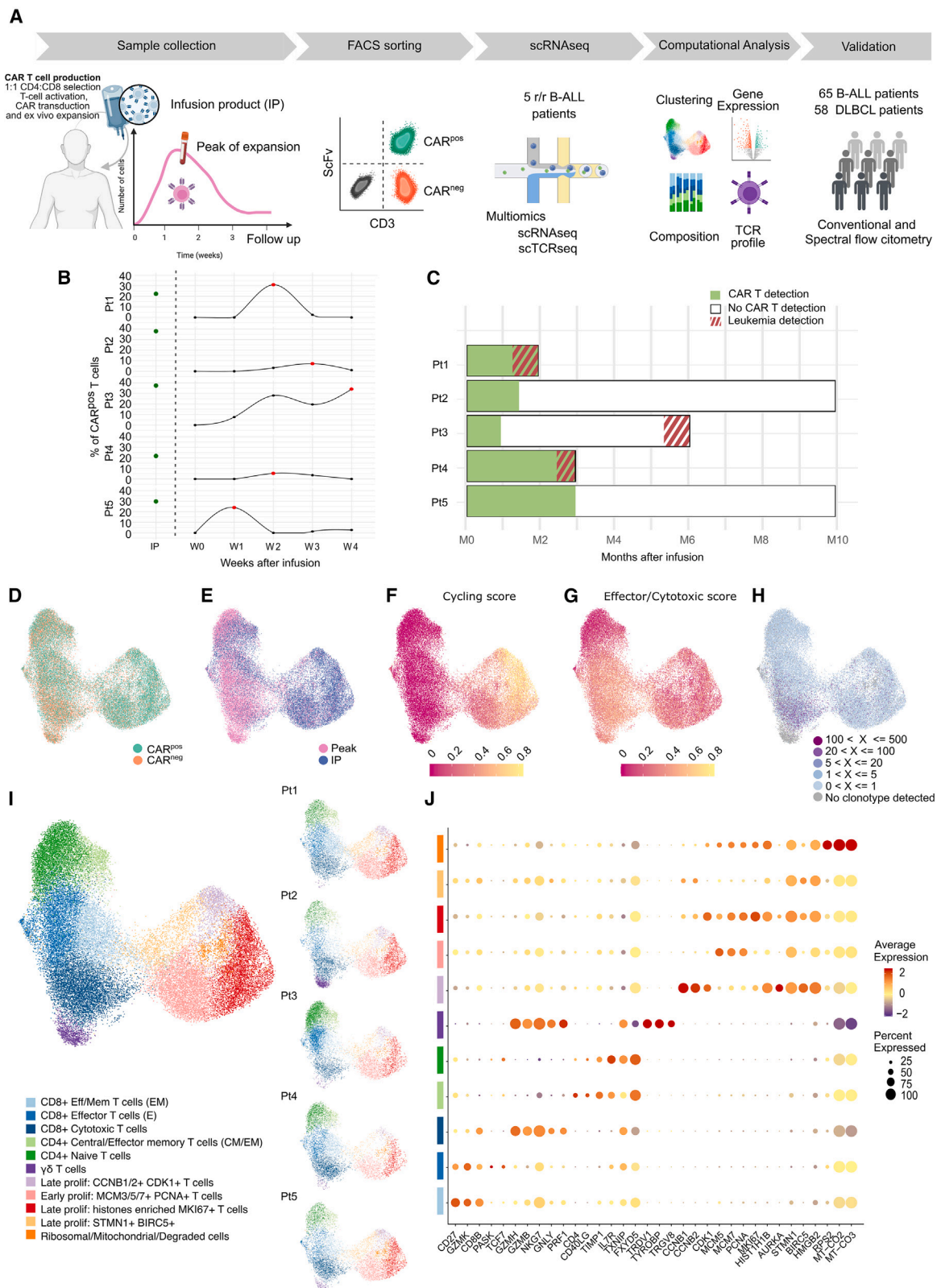
The impact of phenotypic, clonal, and functional heterogeneity of chimeric antigen receptor (CAR)-T cells on clinical outcome remains understudied. Here, we integrate clonal kinetics with transcriptomic heterogeneity resolved by single-cell omics to interrogate cellular dynamics of non-transduced (CAR^{neg}) and transduced (CAR^{pos}) T cells, in the infusion product (IP) and at the CAR-T cell expansion peak in five B cell acute lymphoblastic leukemia (B-ALL) patients treated with CD19CAR-T cells (varni-cel). We identify significant differences in cellular dynamics in response to therapy. CAR^{pos} T cells at IP of complete response patients exhibit a significantly higher CD4:CD8 ratio, validated in a larger cohort B-ALL patients ($n = 47$). Conversely, at the expansion peak, there is a clonal expansion of CD8⁺ effector memory and cytotoxic T cells. Cytotoxic CAR^{pos} $\gamma\delta$ -T cells expansion correlates with treatment efficacy validated in a cohort of B-ALL ($n = 18$) and diffuse large B cell lymphoma (DLBCL) patients ($n = 58$). Our data provide insights into the complexity of T cell responses following CAR-T cell therapy and suggest drivers of immunotherapy response.

INTRODUCTION

Chimeric antigen receptor (CAR)-T cell therapy has emerged as a groundbreaking treatment option for relapsed/refractory (r/r) hematological malignancies, particularly B cell acute lymphoblastic leukemia (B-ALL), B cell lymphoma, and multiple

myeloma. Impressive clinical complete response (CR) rates are achieved after infusion of CD19-directed CAR-T (CD19CAR-T) cells^{1–5} for B-ALL and lymphoma, and BCMA-directed CAR-T cells⁶ for multiple myeloma. Unfortunately, sustained CR rates at one year following treatment have been observed in <50% of these patients, indicating a pressing need to better





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understand the underlying cellular and molecular features of CAR-T cell therapy leading to different patient outcomes.^{7,8}

The therapeutic performance of CAR-T cells depends on multiple factors, including the design of the CAR construct, antigen density, the quality and starting phenotype of the T cells sourced for manufacturing, the lymphodepletion and infusion scheme, the tumor burden, and the tumor microenvironment.^{9–15} Additionally, clinico-biological factors that confer immune resistance to the tumor cells are key determinants of clinical response and durability, but the identification of these factors that predict clinical outcomes remains challenging.

Recent advances in single-cell multi-omics approaches have provided new insights into the clonal kinetics and heterogeneity of CAR-T cell products, enabling the identification and characterization of CAR-T cell populations associated with sustained remission.^{16–22} However, these studies have also described variability in the clonal composition and transcriptional signatures of CAR-T cells and how this translates into the distinct functional T cell subsets, highlighting the need for additional studies to identify those T cell subtypes/clones responsible for short-term efficacy and long-term persistence.^{11,13}

Here, we integrated clonal kinetics using single-cell $\alpha\beta$ T cell receptor (TCR) sequencing (sc- $\alpha\beta$ TCR-seq) with transcriptomic heterogeneity resolved by single-cell RNA sequencing (scRNA-seq) to explore the cellular dynamics response of both non-transduced (CAR^{neg}) and transduced (CAR^{pos}) T cells purified by flow cytometry based on CAR expression. We examined both (CAR^{neg} and CAR^{pos}) T cell fractions in the same manufactured infusion products (IPs) and in peripheral blood (PB) samples at the time of CAR-T cell expansion peak following infusion (peak) in 5 adult r/r B-ALL patients treated with the CD19CAR-T product varnimcabtagene autoleucel (varni-cel).^{8,23,24} Our data reveal relevant differences in the cellular dynamics between CAR^{pos} and CAR^{neg} T cells in response to varni-cel therapy. Expanded cytotoxic CAR^{pos} $\gamma\delta$ T cells, not traceable by $\alpha\beta$ TCR, emerged at the expansion peak in all patients, and the extent of *in vivo* expansion of CAR^{pos} $\gamma\delta$ T cells correlated with treatment efficacy. Clinico-biological associations were validated in larger cohorts of CD19CAR-T-treated B-ALL (varni-cel) and diffuse large B cell lymphoma (DLBCL; either lisa-cel or axi-cel) patients. This study highlights the complexity and diversity of the T cell response following CD19CAR-T cell therapy, which may not solely depend on CAR^{pos} $\alpha\beta$ T cells.

RESULTS

Multimodal identification of T cell subtypes in response to CD19CAR-T cell therapy

To investigate immune responses to CD19CAR-T cell therapy and understand the impact of CAR expression on T cell identities, we analyzed the single-cell transcriptional and immune repertoire profiles of transduced (CAR^{pos}) and non-transduced (CAR^{neg}) T cells in the IP and at peak expansion in 5 patients with r/r B-ALL who received the CD19CAR-T product varni-cel.^{8,23,24} (Figures 1A and S1A). Clinico-biological information of the patients from the discovery cohort is shown in Table S1 (further information of the validation cohorts in Table S2). Varni-cel was produced and infused as detailed elsewhere.²⁵ CAR-T cell transduction levels in the final IP ranged between 23% and 38%, and the infused CAR-T cells exhibited different degrees of expansion in patients, with peak expansion occurring between week 1 and 4 post-infusion (Figure 1B). Patient outcomes were monitored for up to 10 months (Figure 1C). Patients 1 and 4 experienced early relapse with remission periods of 2 and 3 months, respectively. During their relapse, both CAR^{pos} T cells and leukemic cells coexisted. Patients 2 and 5 showed a sustained CR, whereas patient 3 experienced a relapse at month 6 post-infusion.

A total of 37,100 high-quality single T cells were analyzed across all samples and time points. The number of sequenced cells was consistent across the different experimental conditions (Tables S3 and S4). CAR transcripts were found to be evenly distributed among all patients in both IP and peak samples (Figures S1B and S1C). Notably, CAR transcripts were minimally detected in CAR^{neg} T cells, and any such minimal detections were subsequently filtered out (Figures S1B and S1C; Tables S3 and S4). To identify distinct T cell subpopulations and characterize their transcriptional heterogeneity, we first normalized and scaled the single-cell gene expression counts, and we next performed principal-component analysis (PCA) on the highly variable genes to reduce the dimensionality of the data and maintain only the most biologically informative genes. Then, we used a shared nearest-neighbor graph-based clustering method coupled with uniform manifold approximation and projection (UMAP) to identify and visualize bi-dimensionally cells with similar transcriptional profiles (see STAR Methods). There was a high overlap between cells from the CAR^{pos} and

Figure 1. The landscape of T cell populations in r/r B-ALL patients treated with varni-cel

(A) Schematic overview of the experimental design. CAR-T cells were manufactured from leukapheresis products. CD4/CD8 T cells were selected, activated, transduced with the CD19CAR construct, and expanded *in vitro*. Upon manufacturing, a sample was collected from each patient's product prior to infusion, conforming the "Infusion Product" (IP). Subsequently, patients were monitored weekly to assess the CAR-T cell expansion, and the sample with the highest level of CAR^{pos} T cells over this period conformed the peak of expansion (Peak). CAR^{neg} and CAR^{pos} T cell fractions were fluorescence-activated cell sorting-purified from IP and peak samples and prepared in single-cell suspensions for scRNA-seq and scTCR-seq.

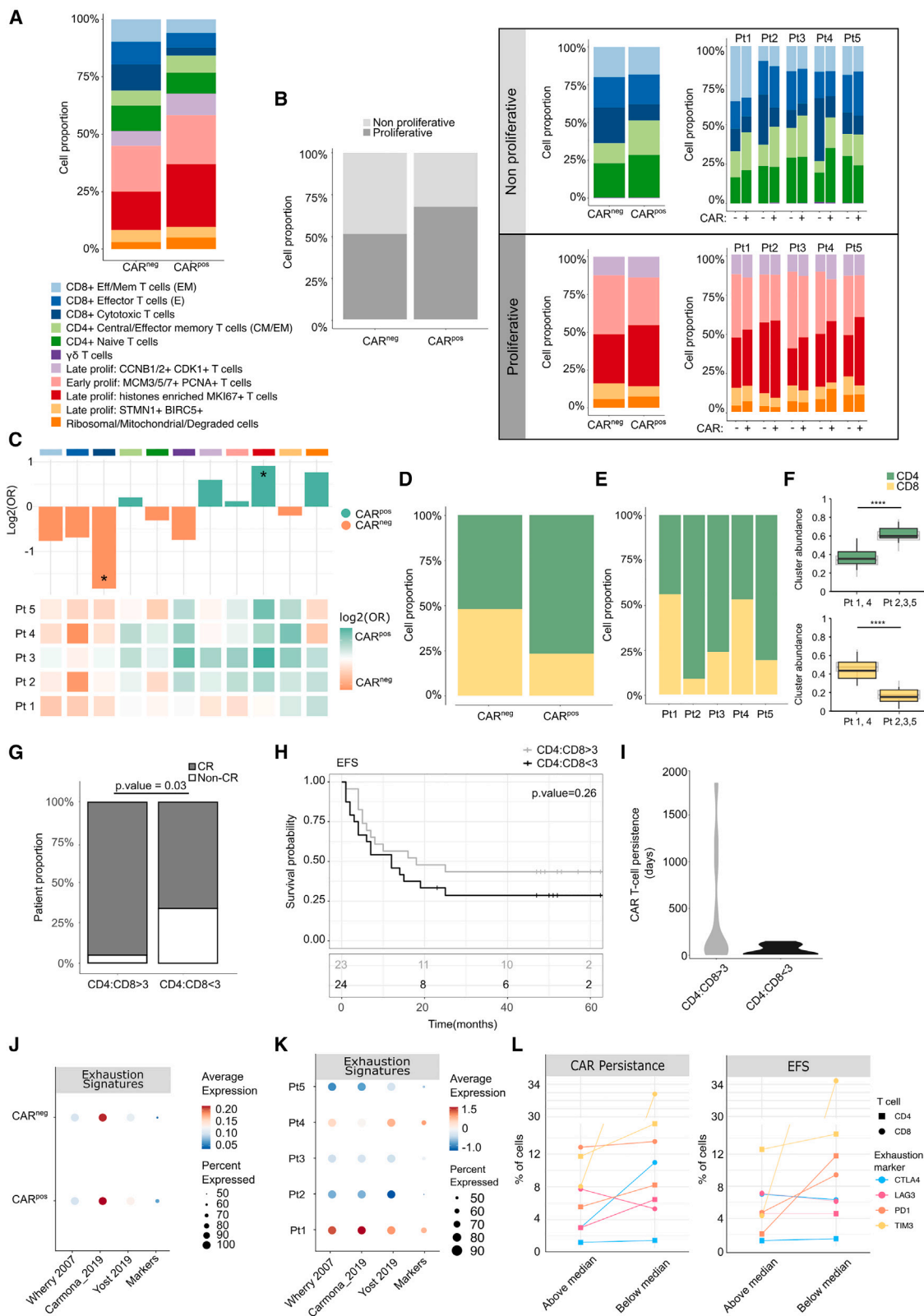
(B) Flow cytometry monitoring for CAR^{pos} T cell expansion over the first four weeks after infusion. The percentage of CAR^{pos} T cells in the IP is indicated as a green dot whereas peak samples are represented as red dots.

(C) Summary of the clinical evolution of the patients over a ten-month follow-up. Solid green and white bars indicate the presence or absence (no detection) of CAR^{pos} T cells in disease-free patients, respectively. Stripped red lines indicate patient relapse.

(D–H) Uniform manifold approximation and projection (UMAP) of the 37,100 cells pooled from all the samples from the 5 patients colored as follows: (D) CAR^{pos} vs. CAR^{neg} cells and (E) IP vs. peak, (F and G) scaled from high (yellow) to low expression (magenta) of cycling score (F), and effector/cytotoxic score (G). (H) UMAP representing clonotype sizes (from gray to purple).

(I) UMAP colored by the 11 different cell clusters identified by unsupervised clustering both combined (left) and separated by patient (right).

(J) Dot plot of the top gene markers by cluster identified by differential expression analysis.



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CAR^{neg} subpopulations (Figure 1D), while T cells tended to be more distinct when based on their sampling time (Figure 1E). Specifically, the peak samples exhibited a higher abundance of non-cycling clonally expanded effector/cytotoxic T cells when compared with the IP samples (Figures 1E–1H). The analysis of cluster-specific gene expression using those genes differentially expressed (DEGs) among clusters allowed us to identify 11 subtypes of T cells, encompassing a wide range of differentiation states, from the less mature and proliferating T cells to the most differentiated CD8⁺ effector memory and cytotoxic T cells (Figures 1I, 1J, S1D, and S1E).

Four clusters of highly proliferative cells were identified based on the expression of genes involved in cell cycling, DNA replication, and proliferation, including cyclins/cyclin-dependent kinases, HMGB proteins, PCNA, MCM5, MCM7, MKI67, and AURKA (Figures 1I and 1J). These clusters also exhibited a high cycling score (Figures 1F and 1J). Additionally, a CD4⁺ naive T cell cluster was identified that expressed both CD4 and IL7R, with low cycling and effector cytotoxic scores. Central effector memory markers were expressed in the remaining clusters, including one CD4⁺ and three CD8⁺ T cell clusters (Figures 1I and 1J). One CD8⁺ T cell cluster contained effector cytotoxic markers, including GZMH, GZMB, GNLY, NKG7, and PRF1, whereas the others CD8⁺ effector clusters expressed GZMK, indicative of a more effector profile, as previously described.^{18,26} Furthermore, the gradient expression of CD27 in these CD8⁺ T cells clearly reflects their differentiation and functional diversity within the immune response.

Of note, a distinct cluster of cytotoxic $\gamma\delta$ T cells expressing the $\gamma\delta$ T cell markers TCR delta variable 1 (TRDV1) and TCR gamma variable 8 (TRGV8) and the cytotoxic markers GNLY, NKG7, GZMH, and GZMB but lacking $\alpha\beta$ TCRs was identified (Figures 1G–1J). While all T cell subtypes were present in each patient, the Morisita index revealed no shared TCR sequences across patients, indicating a lack of similarity between their TCR repertoires, highlighting the vast diversity of individual im-

mune responses (Figure S1F). In contrast, we observed some overlap in the TCR repertoires between the CAR^{pos} and CAR^{neg} subpopulations within the same patient (Figure S1F). Overall, these findings provide a comprehensive characterization of the different T cell populations and their molecular profiles, which can aid in understanding the CD19CAR-T cell response.

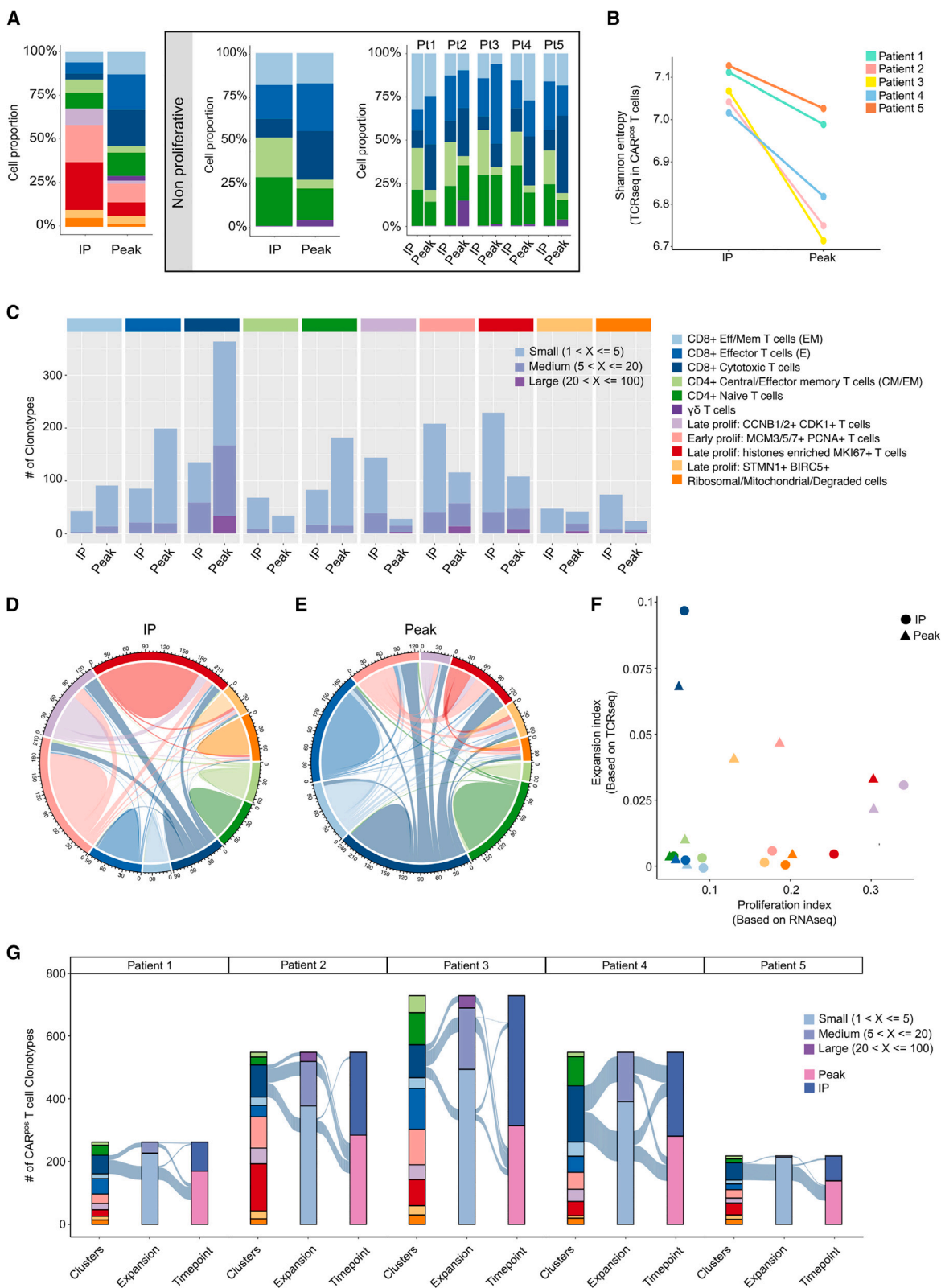
Changes in subtype composition between CAR^{pos} and CAR^{neg} T cells in the IP link to treatment outcome

To understand the impact of CAR transduction during manufacturing, we compared T cell subtype compositions between CAR^{pos} and CAR^{neg} T cells at the IP (Figure 2A). The proliferation was the primary source of variation upon manufacturing, with a higher number of proliferative cells found within the CAR^{pos} T cell population (Figure 2B). This observation was further validated at the transcriptomic level, where differential gene expression between CAR^{pos} and CAR^{neg} T cells in the IP suggested an increased proliferative activity in CAR^{pos} T cells (Figure S2A). However, when examining proliferative and non-proliferative cells separately, we observed a consistent cell subtype composition of proliferative cells in all patients (Figure 2B). By contrast, a meaningful shift in the abundance of CD8⁺ T cells was observed in non-proliferating cells, with CAR^{neg} T cells showing a significant higher proportion of CD8⁺ effector cytotoxic and memory T cells (Figures 2B and 2C).

To gain further insight into the composition of the T cell populations and their potential functional implications following CAR transduction, we assigned a CD4 or CD8 phenotype to each cell by projecting the cells onto different publicly available single-cell multi-omic references (see STAR Methods) (Figure S2B). We found a consistent and substantial shift in the abundance of CD4⁺ T cells in CAR^{pos} products across all patients (Figures 2D and S2C), which may be experimentally explained by the biased CAR-expressing lentiviral transduction of CD4⁺ over CD8⁺ T cells ($n = 147$ T cell donors; Figure S2D). This higher

Figure 2. Differences among CAR^{pos} and CAR^{neg} T cells in the IP

- (A) Stacked bar plot showing T cell subtype composition of CAR^{pos} and CAR^{neg} T cells in the IP.
- (B) Left, proportion of proliferative and non-proliferative CAR^{pos} and CAR^{neg} T cells in the IP. Right, stacked bar plots showing the T cell subtype composition within the non-proliferative (upper panels) and proliferative (lower panels) CAR^{pos} and CAR^{neg} T cells both overall and by patient in the IP.
- (C) Barplot (overall) and heatmap (by patient) showing the differential abundance analysis of CAR^{pos} and CAR^{neg} T cell clusters in the IP. Cluster abundance is represented as the log2 odd ratio between CAR^{pos} and CAR^{neg} subsets. Significant differences are computed with sccomp (see STAR Methods) and are indicated with * ($p \leq 0.05$).
- (D and E) Distribution of predicted CD4 and CD8 T cell classifications across all cells in the IP (D) and CAR^{pos} CD4 and CD8 T cell classifications, broken down by individual patient, showing how the CAR^{pos} CD4:CD8 T cell ratio varies among patients (E).
- (F) Boxplots depicting the differential abundance of CAR^{pos} CD4 and CD8 T cell populations in the IP across different patients, grouped by clinical outcome: patients 1 and 4 show early relapses and patients 2, 3, and 5 are long-term responders. Significant differences are computed with sccomp (see STAR Methods) and are indicated with **** p values ≤ 0.0001 .
- (G) In a retrospective validation cohort of 47 patients with r/r B-ALL treated with Varni-cel, the CR rates were compared between two groups based on a CAR^{pos} CD4:CD8 ratio >3 or <3 (chi-squared test; p value = 0.03; $n = 16$ and $n = 44$).
- (H) Kaplan-Meier EFS curve for r/r B-ALL patients who received varni-cel products with a CAR^{pos} CD4:CD8 ratio >3 (gray line; $n = 16$) and <3 (black line; $n = 44$), respectively (Log rank test; p value = 0.26). Time is represented in months.
- (I) Violin plot showing the CAR-T cell persistence, clinically defined as loss of B cell aplasia, for varni-cel-treated r/r B-ALL patients who received products with a CAR^{pos} CD4:CD8 ratio >3 or <3 ($n = 14$ and $n = 11$; Mann-Whitney test; p value = 0.059).
- (J and K) Dot plots showing the expression of four exhaustion signatures in CAR^{pos} and CAR^{neg} T cells at the IP both overall (J) and by patient (K). The Ucell method was used to score each signature (Table S5).
- (L) The mean percentage of CAR-T cells expressing the indicated exhaustion markers at the IP was retrospectively compared in varni-cel-treated r/r B-ALL patients categorized based on whether their CAR-T cell persistence (left panel) or EFS (right panel) was above or below the median. Squares and circles correspond to CD4⁺ and CD8⁺ T cells, respectively.



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representation of CD4⁺ T cells in the CAR^{pos} fraction of the IP was independent of CD4 T cell-specific activation or tonic signaling (Figures S2E and S2F; Table S5). Of note, a significant diminished proportion of CAR^{pos} CD4⁺ T cells in the IP was associated with early relapses (Figures 2E, 2F, and 1C). This was further confirmed by flow cytometry in a retrospective validation cohort ($n = 47$) of varni-cel-treated r/r B-ALL patients (Table S2). In this expanded cohort, 95% and 60% of patients infused with a CAR^{pos} CD4:CD8 ratio >3 or <3, respectively, achieved CR after treatment ($p = 0.03$) (Figures 2G and S2G). Accordingly, the five-year event-free survival (EFS) was superior in patients who had received a CAR-T cell product with a CAR^{pos} CD4:CD8 ratio >3 (43.5% [confidence interval (CI) 27.3%–69.3%] vs. 28.5% [CI 15%–54.3%]) (Figure 2H). Of note, a trend toward higher CAR persistence was observed in patients who received a CAR-T cell product with a CAR^{pos} CD4:CD8 ratio >3 (mean CAR persistence 493 [CI 57–1883] vs. 106 [CI 56–196]), p value = 0.059 (Figure 2I).

Additionally, we assessed relevant transcriptional signatures related to T cell exhaustion that might affect the clinical outcome of the CAR-T cell therapy (Table S5) and found no significant differences between CAR^{pos} and CAR^{neg} T cell populations in the IP (Figure 2J). However, patients with early relapses had consistently higher scores of different exhaustion signatures, irrespective of the expression of the CAR construct (Figures 2K and S2H; Table S5), indicating a CAR-independent exhausted T cell response in these patients. We next set out to validate T cell exhaustion as a biomarker predictive of patient outcomes following CAR-T cell therapy in an independent validation cohort of varni-cel-treated r/r B-ALL patients ($n = 47$). To this end, we analyzed the expression of major immune checkpoint molecules (CTLA4, LAG3, PD1, and TIM3) by flow cytometry, separately in CD4⁺ and CD8⁺ T cells at the IP. We then correlated these expression levels with CAR-T cell persistence and EFS. We found an inverse trend in the correlation between the expression of T cell exhaustion molecules and both the persistence of CAR-T cells and EFS, categorized as below or above the median. This was especially significant for the levels of CD8⁺TIM3⁺ T cells (p value = 0.0001) (Figure 2L).

Transcriptomic and clonal dynamics of CAR^{pos} T cells from ex vivo manufacturing (IP) to in vivo expansion peak

To gain a more comprehensive understanding of the CAR-T cell response dynamics, we investigated the evolution of CAR^{pos} T cells at both the transcriptional and clonal levels. By comparing

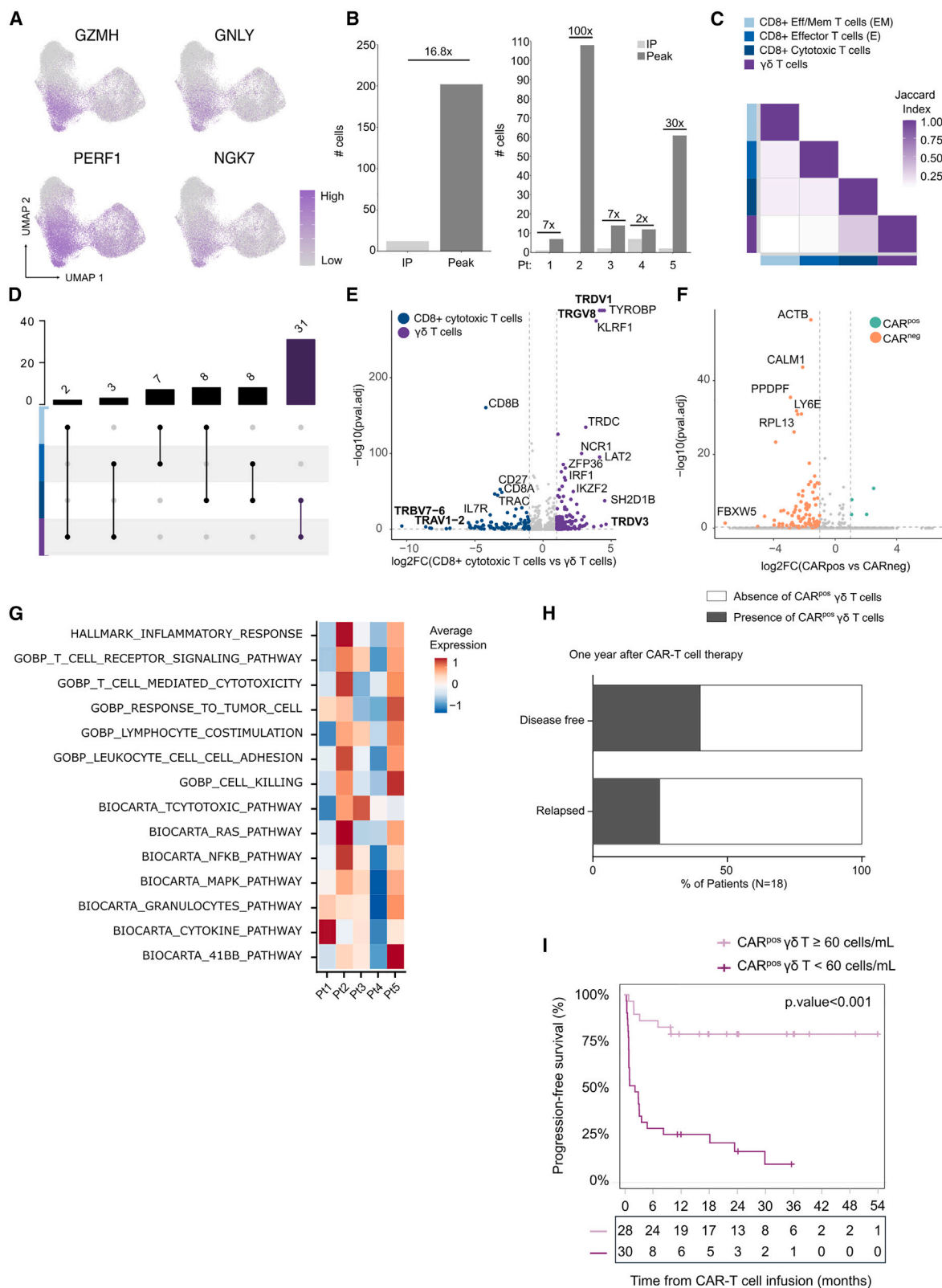
changes in T cell composition from the IP to the expansion peak in the discovery cohort, we identified a significant decline in proliferative cells, which were subsequently replaced by CD8⁺ T cells (Figure 3A), a phenomenon observed also for the CAR^{neg} T cells (Figures S3A and S3B). Within the non-proliferative CAR^{pos} T cells, an inverted trend in the CD4:CD8 ratio was observed at the expansion peak in all patients, with CD8⁺ T subtypes representing 75% of the non-proliferative cellular product (Figure 3A). Of note, a population of CAR^{pos} and CAR^{neg} $\gamma\delta$ T cells emerged significantly at the expansion peak in all patients (Figures 3A, S3A, and S3B).

To better explore the complexity of the CAR-T cell responses and quantify the diversity of TCR repertoires, we employed the Shannon entropy index, a diversity measure that considers both abundance and distribution of TCR sequences. Comparing the Shannon entropy indices between IP and peak revealed a consistent trend indicative of an *in vivo* clonal expansion of CAR^{pos} T cell clones over time in all the patients studied (Figure 3B). This expansion of T cell clones was particularly pronounced within the CD8⁺ T cell fraction, with the CD8⁺ cytotoxic subtype showing the largest number of clonotypes and magnitude of expansion (Figures 3C and 1H), a phenomenon observed in both CAR^{pos} and CAR^{neg} CD8⁺ T cells (Figure S3C).

Notably, the process of clonal expansion and diversification of the CD8 cytotoxic cells seemed to be continuous, occurring from the IP to the peak. As the process unfolded, the emergence of shared clones between CD8⁺ cytotoxic T cells and other proliferative T cells increased at the expansion peak, indicating the ongoing diversification of the T cell response (Figures 3D, 3E, and S3D). The transcriptional regulation of this continuous process explored through pseudotime analysis highlights the dynamic changes of the cell states from the IP to the peak (Figure S3E) and identifies the genes most differentially expressed as a function of (pseudo)time (Figure S3F). T cell states with a more proliferative transcriptomic profile were predominantly found at the beginning of pseudotime, mostly consisting of IP samples. In contrast, T cell states characterized by effector and cytotoxic profiles were positioned at the end of the trajectory, largely comprising peak T cells, reflecting the continuous process of T cell expansion (Figures S3E and S3F). It is possible that the high expansion of CD8⁺ T cells provides a larger pool of cells available for differentiation and recruitment into other T cell subsets. This is supported by the finding that CAR^{pos} CD8⁺ cytotoxic T cells have a high expansion score but a low proliferative score (Figure 3F), suggesting that their increase in number is not

Figure 3. Longitudinal analysis of CAR^{pos} T cells across time points (IP and peak)

- Left, stacked barplots of CAR^{pos} T cell subtype composition at the IP and peak. Right, stacked barplots of CAR^{pos} T cell subtype composition analyzed within the non-proliferative T cells at the IP and peak displayed both overall and by individual patient.
- IP-peak paired Shannon entropy measure calculated for each patient to evaluate the diversity of the TCR repertoire within the CAR^{pos} T cells.
- Occupied TCR repertoire space for CAR^{pos} T cells by cell type/cluster and time point (IP and peak). Only small ($1 < x \leq 5$), medium ($5 < x \leq 20$), and large ($20 < x \leq 100$) clonotypes were considered.
- D and E) Chord diagrams showing the interconnection of shared expanded clonotypes among the different T cell clusters at the IP (D) and peak (E).
- Scatterplot showing the clonal expansion and the proliferation indexes of the CAR^{pos} T cells at IP (circles) and peak (triangles). Colors indicate the T cell cluster. Proliferation index was calculated based on scRNA-seq data, and expansion index was computed by Startrac software from scTCR-seq data (see STAR Methods).
- Alluvial plot depicting the total number of CAR^{pos} T cell clonotypes per patient. Each bar is color-coded to show the proportion of the different T cell subtypes (clusters), as well as T cell expansion and sample collection time points per patient. Blue lines connect CD8⁺ cytotoxic T cell clonotypes to their corresponding clonal expansions and sample collection time points.



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due to cell division but instead to cell survival, recruitment, or differentiation. By contrast, CAR^{pos} CD4⁺ T cell subtypes exhibited the lowest expansion and proliferation scores, indicating that CAR-T cell responses are characterized by the preferential expansion and survival of specific T cell subsets, rather than the global expansion of all T cell populations.

We further investigated the T cell response following CAR-T cell therapy by comparing the expansion and clonal diversity among patients with varying clinical responses. This analysis revealed differences in T cell expansion and clonal diversity between patients (Figures 3G, S3G, and S3H). We observed a consistent expansion of CD8⁺ cytotoxic T cells in all patients at the peak, in both CAR^{pos} (Figure 3G) and CAR^{neg} fractions (Figure S3G), except for patient 4 where the most expanded clones were already observed at IP. However, no correlation was found between the number of shared clones or their rate of expansion and treatment response (Figures 3G, 1C, S3G, and S3H). Overall, these findings suggest that the expansion of CD8⁺ cytotoxic T cells may play a major role in determining the response to CAR-T cell therapy, but other factors may also be involved. In addition, the difference in the patterns of clonal expansion and sharing between patients may reflect differences in the underlying mechanisms of CAR-T cell response and the complexity of the T cell repertoire.

***In vivo* expansion of CAR^{pos} γ δ T cells links to a more favorable treatment outcome**

We initially identified a population of CAR^{pos} T cells characterized by high cytotoxic activity (Figure 1G) and undetectable $\alpha\beta$ TCR expression (Figure 1H). We determined that these cells are cytotoxic $\gamma\delta$ T cells based on their expression of the $\gamma\delta$ TCR markers TRDV1 and TRDV8 and cytotoxic genes, such as granzymes, GZMH, PRF1, GNLY, and NKG7 (Figures 4A and 1J). The absolute number of CAR^{pos} $\gamma\delta$ T cells was very low upon manufacture (IP) but was highly expanded *in vivo* (between 2- and 100-fold) at the expansion peak (Figure 4B). Remarkably, the two patients in the discovery cohort with a massive *in vivo* expansion of CAR^{pos} $\gamma\delta$ T cells (patient2 and patient5) showed a more sustained clinical response (Figures 4B and 1C). This trend was also observed for CAR^{neg} $\gamma\delta$ T cells (Figure S3A).

To characterize the transcriptomic differences between these cytotoxic $\gamma\delta$ T cells and the CD8⁺ T cell clusters, we calculated the Jaccard index, which reflects similarity, on the top 100 DEGs of each T cell subtype. The highest Jaccard index was

observed between cytotoxic CD8⁺ T cells and $\gamma\delta$ T cells (Figure 4C), with 31 shared genes identified (Figure 4D). The principal differences between these two clusters were primarily associated with the TCR chain genes. Specifically, TRDV1, TRDV3, and TRGV8 were expressed in $\gamma\delta$ T cells, whereas TRAV1-2 and TRBV7-6 were expressed in CD8⁺ cytotoxic T cells (Figure 4E). Of note, there were not substantial transcriptomic differences between CAR^{pos} and CAR^{neg} $\gamma\delta$ T cells (Figure 4F).

Given the observed variability in the $\gamma\delta$ T cells abundance and their known high cytotoxic potential in therapeutic context, we further investigated possible differences in cytotoxicity at the transcriptomic level between patient clinical outcomes by examining the expression of gene sets related to CAR-T cell cytotoxic mechanisms.^{27–29} The cytotoxicity was notably higher in $\gamma\delta$ T cells from patients 2 and 5, underscoring the anti-tumoral activity of both CAR-expressing and steady-state $\gamma\delta$ T cells that were *ex vivo* activated during the manufacturing process (Figure 4G).

The clinical relevance of $\gamma\delta$ T cells was further validated by conventional and spectral flow cytometry in two retrospective larger cohorts of CD19CAR-T cell-treated r/r B-ALL and DLBCL patients (Table S2). In the context of B-ALL ($n = 18$), the data suggest a correlation between the presence of CAR^{pos} $\gamma\delta$ T cells and patient outcomes one-year post-treatment with varni-cel. Specifically, among patients who remained disease free, 40% (4 out of 10 patients) had persisting CAR^{pos} $\gamma\delta$ T cells whereas only 25% (2 out of 8 patients) of patients who experienced a relapse had detectable CAR^{pos} $\gamma\delta$ T cells one-year after treatment (Figure 4H). This finding was also confirmed in an independent cohort of r/r DLBCL patients. In this cohort of 58 patients treated with either axi-cel ($n = 33$) or tisa-cel ($n = 25$), those with CAR^{pos} $\gamma\delta$ T cell counts of >60 cells/mL at the peak of cell expansion showed significantly better progression-free survival ($p < 0.001$, Figures 4I and S1A; Table S2). Collectively, these results underscore the crucial role that CAR^{pos} $\gamma\delta$ T cells play in the clinical response, indicating their potential as biomarkers for predicting clinical outcomes.

DISCUSSION

CAR-T cell therapy has emerged as an effective and safe therapeutic approach in patients with r/r hematological malignancies. Our biological understanding and clinical management of such an adoptive immunotherapy are still in their early stages, but it is undoubtedly poised to become a therapeutic intervention

Figure 4. *In vivo* expansion of CAR^{pos} $\gamma\delta$ T cells correlates with treatment outcome

- UMAP plots color-coded by the expression of the indicated cytotoxic genes.
- Bar plots showing the absolute number of the identified CAR^{pos} $\gamma\delta$ T cells at the IP (light gray) and peak (gray), combining all samples (left) and separating by patient (right).
- Similarity Jaccard index comparing CD8⁺ and $\gamma\delta$ T cell clusters. The index was computed considering the top 100 DEGs from each cluster.
- Upset plot showing the number of shared markers between CD8⁺ and $\gamma\delta$ T cell clusters, considering the top 100 DEGs from each cluster.
- Volcano plot depicting the differential expression analysis between $\gamma\delta$ T cells and CD8⁺ cytotoxic T cells at the peak of expansion (E) and between CAR^{pos} and CAR^{neg} $\gamma\delta$ T cells at the peak of expansion (F).
- Heatmap representing the gene expression signatures related to CAR-T cell cytotoxicity in $\gamma\delta$ T cells at the peak of expansion, separated by patient. The Ucell method was used to score each signature (see STAR Methods).
- Proportion of disease-free and relapsed varni-cel-treated B-ALL patients with persisting CAR^{pos} $\gamma\delta$ T cells detectable one year after treatment assessed by flow cytometry in the retrospective validation cohort ($n = 18$; $n = 10$ and $n = 8$).
- Prognostic impact of the number of CAR^{pos} $\gamma\delta$ T cells at the IP on 5-year progression-free survival in a validation cohort of 58 r/r DLBCL patients treated with axi-cel ($n = 33$) and lisa-cel ($n = 25$) products (high $\gamma\delta$ T cells $n = 28$ and low $\gamma\delta$ T cells $n = 30$; Log rank test; p value < 0.001).

during earlier stages of the disease. CAR-T cell therapy will eventually benefit many more patients and even a broader spectrum of tumors once significant logistical and biological milestones are achieved. However, there is a high demand for reliable predictive biomarkers of clinical success, measured by the persistence of CAR-T cells capable of long-term disease control without relapse or immune escape. In this regard, the design of CAR constructs, the target of choice, antigen density, the quality and phenotype of the manufactured T cells, the lymphodepletion and infusion scheme, tumor burden, tumor microenvironment, and other factors that confer immune resistance to tumor cells are key determinants of clinical response.²⁸

To help identifying biomarkers of clinical outcome we have integrated single-cell clonal kinetics with transcriptomic heterogeneity to explore the cellular dynamic responses of both non-transduced (CAR^{neg}) and transduced (CAR^{pos}) T cell fractions longitudinally sampled from the IP and from PB at the time of CAR-T cell expansion peak.²³ Although recent advances in single-cell multi-omics have provided relevant information into the clonal kinetics and heterogeneity of other CAR-T cell products,^{6,30–32} this study aimed to understand the impact of patient-matched CAR^{pos} and CAR^{neg} T cells by assessing their individual and synergistic effects on the clinical response of r/r B-ALL patients treated with varni-cel.^{8,23,25,33} Our cohort included patients who exhibited distinct patterns of clinical response to varni-cel: some patients relapsed despite the presence of detectable CAR-T cells, while others experienced prolonged remission without persistence of CAR-T cells. Neither age, gender, molecular/cytogenetic features of the disease, and disease burden nor the number of CAR^{pos} T cells at infusion correlated with clinical outcomes in our discovery cohort. This suggests that either leukemic cell autonomous mechanisms of disease progression or variability in the clonal, phenotypic, or functional composition of the infused T cells may be responsible for patient outcomes.

Our results unravel significant differences in the cellular dynamics between CAR^{pos} and CAR^{neg} T cells and highlight the complexity and diversity of the T cell responses following CD19CAR-T cell therapy. Upon manufacturing, a consistent and substantial shift in the abundance of CD4⁺ T cells was observed in CAR^{pos} T cell fraction across all patients, which was not associated with CD4 T cell-specific proliferation, activation, or tonic signaling, but with a significant higher transduction of CD4⁺ T cells. Importantly, the CAR^{pos} CD4:CD8 ratio in the IP was linked to clinical outcome. A CAR^{pos} CD4:CD8 ratio <3 was associated with lower CR rates, shorter CAR-T cell persistence, and lower EFS as validated by flow cytometry in a retrospective validation cohort of 47 patients with r/r B-ALL treated with varni-cel. Of note, although no differences in T cell exhaustion, activation, or tonic signaling signatures could be found between CAR^{pos} and CAR^{neg} T cells in the IP, refractory patients had consistently higher scores of different exhaustion signatures, irrespective of the expression of the CAR construct, indicating that a CAR-independent T cell dysfunction could potentially predict patient outcomes following CAR-T cell therapy. In an independent validation cohort of varni-cel-treated r/r B-ALL patients (*n* = 47), we observed an inverse correlation between the expression of T cell exhaustion markers, particularly CD8⁺TIM3⁺ T cells,

and both CAR-T cell persistence and EFS. This finding further supports T cell exhaustion as a predictive biomarker for patient outcomes. Despite ongoing controversy regarding the reliability of these markers as predictors of clinical response,²⁷ our findings align with previous studies suggesting that the CD4:CD8 T cell ratio^{34–36} and exhaustion T cell signatures^{20,21,37–39} are useful predictors of clinical response.^{13,17,40,41} It might thus be possible to identify those patients with a higher risk for relapse or who might benefit from additional therapies to enhance the durability/persistence of the T cell response. Additionally, these data provide clues about the potential advantage of pre-selecting less exhausted T cell populations, for CAR-T cell manufacturing.

The expression of the CAR serves as a label/identifier to longitudinally trace the cellular dynamics, proliferation, expansion, and clonality of CAR-T cells after infusion in the patient. Although it is not possible to unequivocally determine if some CAR^{neg} T cells were initially present at the IP or if they were all newly generated in the patient, we also considered them in our analysis. Integrated single-cell clonal kinetics with transcriptomic heterogeneity revealed a consistently higher abundance of non-cycling clonally expanded effector cytotoxic CD8⁺ T cells in the peak samples. Of note, the CD8⁺ T cell clusters express different granzymes and cytotoxic genes, indicating the existence of distinct effector memory CD8⁺ T cells, as previously reported.^{18,26} Overall, although clonal expansion and TCR diversification of the CD8⁺ T cells seemed to be continuous from the IP to the peak, both CAR^{pos} and CAR^{neg} T cells showed an increased emergence of shared clones between CD8⁺ cytotoxic cells and other proliferative T cells at the expansion peak, highlighting the ongoing diversification of the T cell response in a CAR-independent fashion. The high expansion score but low proliferative score of CD8⁺ cytotoxic T cells suggests that these expanded CD8⁺ T cells have a survival advantage and may serve as a large reservoir of cells available for differentiation into and recruitment of other T cell subsets. By contrast, CAR^{pos} CD4⁺ T cell subtypes exhibited the lowest expansion and proliferation scores, indicating that a preferential expansion and survival of specific T cell subsets, rather than the global expansion of all T cell populations, underlie CAR-T cell responses. However, the lack of TCR repertoire similarity observed across patients coupled to the similarities observed between CAR^{pos} and CAR^{neg} fractions of individual patients underscores the personalized nature of T cell responses. This heterogeneity reflects both the unique antigenic history and the lymphocyte pool of each patient, influencing clonal selection and expansion.

Integrated scTCR-seq and scRNA-seq data also uncovered an independent cluster of cytotoxic T cells expressing multiple GZM genes, PRF1, GNLY, and NKG7, which could not be sequenced through standard $\alpha\beta$ TCR sequencing. These T cells, exhibiting transcriptomic similarities with CD8⁺ cytotoxic T cells, were identified as $\gamma\delta$ T cells based on their expression of $\gamma\delta$ T cell-specific genes, TRDV1 and TRDV8. The absolute number of CAR^{pos} $\gamma\delta$ T cells was very low at the IP but increased significantly *in vivo* (up to 100-fold) at the expansion peak. Intriguingly, a massive *in vivo* expansion of CAR^{pos} $\gamma\delta$ T cells was observed in the long-term responders of the discovery cohort. This was validated by flow cytometry in

retrospective validation cohorts of patients with r/r B-ALL treated with varni-cel and patients with r/r DLBCL treated with tisa-cel or axi-cel, as previously described.^{16,42} Notably, the significant *in vivo* expansion of $\gamma\delta$ T cells after CAR-T cell treatment is particularly striking given that they represent a minor T cell subset in PB and that the CAR-T cell manufacturing process includes the selection of CD4 and CD8 cells, thus depleting >70% of $\gamma\delta$ T cells, which are CD4⁺ and CD8⁺.⁴³ The relevance of $\gamma\delta$ T cells in CAR-T immunotherapies may be due not only to their high cytotoxic capacity, even in the absence of human leukocyte antigen-mediated antigen presentation,^{12,13,42} but also to the intrinsic high capacity of $\gamma\delta$ T cells to migrate from blood and infiltrate the bone marrow (BM) and other peripheral tissues where extranodal involvement by lymphoma cells frequently occurs in r/r DLBCL.^{44,45} This highlights the complexity and diversity of the T cell response following CAR-T cell therapy, which may not solely depend on $\alpha\beta$ T cells, and suggests that future CAR-T cell therapy production could benefit from not filtering out the vast majority of CD4⁺CD8⁺ $\gamma\delta$ T cells (and potentially also other innate T and T/natural killer cells), which have previously been reported to play a crucial role,^{46,47} even in a decade-long leukemia remissions after CD19CAR-T cell therapy.¹⁶ In fact, CAR^{pos} $\gamma\delta$ T cells have demonstrated reactivity *in vitro* and *in vivo* against both CD19⁺ and CD19[−] tumor cells,^{16,42} suggesting that CAR^{pos} $\gamma\delta$ T cells might target B cell tumor cells even after antigen loss and retain specificity via their TCR. Notably, an *in silico* study of 18,000 patients with cancer established that a tumor-associated $\gamma\delta$ T cell gene signature also correlated significantly with improved overall survival across 39 different malignancies, further advocating the application of $\gamma\delta$ T cells in cancer therapy.⁴⁸ Further experimental and clinical studies are needed to prospectively assess the clinical impact of $\gamma\delta$ T cells in CAR-T-treated patients with B-ALL and other blood cancers.

Despite the limited number of patients in the discovery cohort, the robustness of our study is reinforced by the convergence of findings from both experimental investigations and clinical validations in larger retrospective cohorts of varni-cel-treated B-ALL patients and lisa-cel/axi-cel-treated DLBCL patients. This supports the association of a higher CAR^{pos} CD4:CD8 T cell ratio at the IP and greater *in vivo* expansion of CAR^{pos} $\gamma\delta$ T cells with more favorable clinical outcomes.

Limitations of the study

The study has several main limitations. First, the small sample size in the discovery cohort, which included only five patients, although provided valuable initial insights into immune responses to CAR-T cell therapy, restricts the generalizability of the findings. A larger cohort would not only help validate these observations across a more diverse population but also capture a broader range of immune responses. This is especially important given the inherent heterogeneity in patient outcomes, which complicates the full understanding of the underlying mechanisms. A larger sample size would also enable the detection of subtle yet significant variations, such as the expansion of specific T cell subtypes or clonal dynamics, which are crucial for improving patient stratification and tailoring CAR-T cell therapies for more sustained clinical responses. A second limitation is the

incomplete characterization of the T cell repertoire, represented only minimally by the TCR clonotypes in each sample. Given the vast diversity of the human T cell repertoire, where millions of unique TCR sequences contribute to immune responses, this limited representation may overlook key variations in T cell clonality. Rare but functionally important T cell clones could be essential for understanding long-term CAR-T cell persistence or resistance mechanisms. A more comprehensive analysis of the $\alpha\beta$ and $\gamma\delta$ TCR repertoire would offer deeper insights into immune dynamics and help design more effective, patient-specific therapies. This, in turn, would enhance the understanding of how T cell diversity influences clinical outcomes. A third limitation lies in the technical-biological challenge of studying the clonal kinetics of CAR^{neg} T cells at the peak of *in vivo* expansion and the impact these CAR^{neg} T cells have on both clinical outcome and expansion of CAR^{pos} T cells. Since peripheral cytopenias are common and prolonged in patients treated with CAR-T cells, the CAR^{neg} T cells studied in the patient's PB at the time of CAR^{pos} T cell peak expansion may be either pre-existing CAR^{neg} T cells from the IP itself or endogenous T cells from the patient that were neither apheresis collected nor manufactured. Finally, it is difficult to predict whether the T cells with exhaustion phenotypes observed *in vivo* after the expansion of CAR^{pos} T cells represent unfit T cells with limited antitumor function or, conversely, therapeutically efficient cells that have entered anergy and exhaustion after executing their antitumor function in primary and secondary lymphoid organs.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Pablo Menéndez (pmenendez@carrerasresearch.org).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- The sequencing data reported in this paper are deposited in the Gene Expression Omnibus (GEO) database with accession number GSE235760. The data are available for public access and downloading from the GEO database, with no restrictions on its use or redistribution.
- All original code has been deposited at GitHub (<https://github.com/Merguerrero/scCARTcells> and https://github.com/mereulab/CAR-T_Figures) and released through Zenodo (<https://doi.org/10.5281/zenodo.13845259> and <https://doi.org/10.5281/zenodo.13847747>).
- Any additional information required to reanalyze the data reported in this work paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

M.G.-M., A.R.-H., J.L.T., D.B.-R., M.E., E.A.G., D.M., L.M.-M., H.H., A.O., E.M., and C.B. performed experiments and interpreted data. A.B., V.O.-M., N.M.-C., A.M.G.-S., S.R., M.J., A.U.-I., and J.D. led the clinical work and patient inclusion and follow-up for varni-cel, axi-cel, and lisa-cel treatments and provided clinical data. E.M., C.B., and P.M. supervised the research and contributed key knowledge, techniques, and reagents. C.B. and P.M. conceived the study and funded the research. All authors actively participated in the manuscript preparation, with specific contributions from M.G.-M., A.R.-H., J.L.T., E.M., C.B., and P.M. All authors have read and agreed to publish the manuscript.

DECLARATION OF INTERESTS

P.M. is a founder of the spin-off OneChain ImmunoTx, which has no connection with the present research. V.O.-M. reports honoraria and/or consulting fees from BMS/Celgene, Novartis, Gilead/Kite, Miltenyi Biomedicine, Pfizer, and Janssen. A.M.G.-S. reports honoraria and/or consulting fees from Roche, BMS/Celgene, Kyowa Kirin, Novartis, Gilead/Kite, Incyte, Lilly, ADC Therapeutics America, Miltenyi, Ideogen, AbbVie, and Sobi.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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REFERENCES

1. Berdeja, J.G., Madduri, D., Usmani, S.Z., Jakubowiak, A., Agha, M., Cohen, A.D., Stewart, A.K., Hari, P., Htut, M., Lesokhin, A., et al. (2021). Cil-tacabtagene autoleucel, a B-cell maturation antigen-directed chimeric antigen receptor T-cell therapy in patients with relapsed or refractory multiple myeloma (CARTITUDE-1): a phase 1b/2 open-label study. *Lancet Lond. Engl.* 398, 314–324. [https://doi.org/10.1016/S0140-6736\(21\)00933-8](https://doi.org/10.1016/S0140-6736(21)00933-8).
2. Munshi, N.C., Anderson, L.D., Shah, N., Madduri, D., Berdeja, J., Lonial, S., Raje, N., Lin, Y., Siegel, D., Oriol, A., et al. (2021). Idecabtagene vicleucel in Relapsed and Refractory Multiple Myeloma. *N. Engl. J. Med.* 384, 705–716. <https://doi.org/10.1056/NEJMoa2024850>.
3. Davila, M.L., Riviere, I., Wang, X., Bartido, S., Park, J., Curran, K., Chung, S.S., Stefanski, J., Borquez-Ojeda, O., Olszewska, M., et al. (2014). Efficacy and toxicity management of 19-28z CAR T cell therapy in B cell acute lymphoblastic leukemia. *Sci. Transl. Med.* 6, 224ra25. <https://doi.org/10.1126/scitranslmed.3008226>.
4. Brentjens, R.J., Riviere, I., Park, J.H., Davila, M.L., Wang, X., Stefanski, J., Taylor, C., Yeh, R., Bartido, S., Borquez-Ojeda, O., et al. (2011). Safety and persistence of adoptively transferred autologous CD19-targeted T cells in patients with relapsed or chemotherapy refractory B-cell leukemias. *Blood* 118, 4817–4828. <https://doi.org/10.1182/blood-2011-04-348540>.
5. Maude, S.L., Laetsch, T.W., Buechner, J., Rives, S., Boyer, M., Bittencourt, H., Bader, P., Verneris, M.R., Stefanski, H.E., Myers, G.D., et al. (2018). Tisagenlecleucel in Children and Young Adults with B-Cell Lymphoblastic Leukemia. *N. Engl. J. Med.* 378, 439–448. <https://doi.org/10.1056/NEJMoa1709866>.
6. Raje, N., Berdeja, J., Lin, Y., Siegel, D., Jagannath, S., Madduri, D., Liedtke, M., Rosenblatt, J., Maus, M.V., Turka, A., et al. (2019). Anti-BCMA CAR T-Cell Therapy bb2121 in Relapsed or Refractory Multiple Myeloma. *N. Engl. J. Med.* 380, 1726–1737. <https://doi.org/10.1056/NEJMoa1817226>.
7. Park, J.H., Riviere, I., Gonen, M., Wang, X., Sénéchal, B., Curran, K.J., Sauter, C., Wang, Y., Santomasso, B., Mead, E., et al. (2018). Long-Term Follow-up of CD19 CAR Therapy in Acute Lymphoblastic Leukemia. *N. Engl. J. Med.* 378, 449–459. <https://doi.org/10.1056/NEJMoa1709919>.
8. Ortiz-Maldonado, V., Rives, S., Español-Rego, M., Alonso-Saladrigues, A., Montoro, M., Magnano, L., Giné, E., Pascal, M., Díaz-Beyá, M., Castella, M., et al. (2021). Factors associated with the clinical outcome of patients with relapsed/refractory CD19⁺ acute lymphoblastic leukemia treated with ARI-0001 CART19-cell therapy. *J. Immunother. Cancer* 9, e003644. <https://doi.org/10.1136/jitc-2021-003644>.
9. Can, I., Cox, M.J., Siegler, E.L., Sakemura, R., and Kenderian, S.S. (2022). Challenges of chimeric antigen receptor T-cell therapy in chronic lymphocytic leukemia: lessons learned. *Exp. Hematol.* 108, 1–7. <https://doi.org/10.1016/j.exphem.2022.02.001>.
10. Rodríguez-Marquez, P., Calleja-Cervantes, M.E., Serrano, G., Oliver-Caldes, A., Palacios-Berraquero, M.L., Martín-Mallo, A., Calviño, C., Español-Rego, M., Ceballos, C., Lozano, T., et al. (2022). CAR density influences antitumoral efficacy of BCMA CAR T cells and correlates with clinical outcome. *Sci. Adv.* 8, eabo0514. <https://doi.org/10.1126/sciadv.abo0514>.
11. Fraietta, J.A., Nobles, C.L., Sammons, M.A., Lundh, S., Carty, S.A., Reich, T.J., Cogdill, A.P., Morrisette, J.J.D., DeNizio, J.E., Reddy, S., et al. (2018). Disruption of TET2 promotes the therapeutic efficacy of CD19-targeted T cells. *Nature* 558, 307–312. <https://doi.org/10.1038/s41586-018-0178-z>.
12. Kowolik, C.M., Topp, M.S., Gonzalez, S., Pfeiffer, T., Olivares, S., Gonzalez, N., Smith, D.D., Forman, S.J., Jensen, M.C., and Cooper, L.J. (2006). CD28 costimulation provided through a CD19-specific chimeric antigen receptor enhances in vivo persistence and antitumor efficacy of adoptively transferred T cells. *Cancer Res.* 66, 10995–11004. <https://doi.org/10.1158/0008-5472.CAN-06-0160>.
13. Turtle, C.J., Hanafi, L.-A., Berger, C., Gooley, T.A., Cherian, S., Hudecek, M., Sommermeyer, D., Melville, K., Pender, B., Budiarto, T.M., et al. (2016). CD19 CAR-T cells of defined CD4⁺:CD8⁺ composition in adult B cell ALL patients. *J. Clin. Invest.* 126, 2123–2138. <https://doi.org/10.1172/JCI85309>.
14. Sommermeyer, D., Hudecek, M., Kosasih, P.L., Gogishvili, T., Maloney, D.G., Turtle, C.J., and Riddell, S.R. (2016). Chimeric antigen receptor-modified T cells derived from defined CD8⁺ and CD4⁺ subsets confer

- superior antitumor reactivity in vivo. *Leukemia* 30, 492–500. <https://doi.org/10.1038/leu.2015.247>.
15. Imai, C., Mihara, K., Andreansky, M., Nicholson, I.C., Pui, C.-H., Geiger, T.L., and Campana, D. (2004). Chimeric receptors with 4-1BB signaling capacity provoke potent cytotoxicity against acute lymphoblastic leukemia. *Leukemia* 18, 676–684. <https://doi.org/10.1038/sj.leu.2403302>.
16. Melenhorst, J.J., Chen, G.M., Wang, M., Porter, D.L., Chen, C., Collins, M.A., Gao, P., Bandyopadhyay, S., Sun, H., Zhao, Z., et al. (2022). Decade-long leukaemia remissions with persistence of CD4+ CAR T cells. *Nature* 602, 503–509. <https://doi.org/10.1038/s41586-021-04390-6>.
17. Bai, Z., Lundh, S., Kim, D., Woodhouse, S., Barrett, D.M., Myers, R.M., Grupp, S.A., Maus, M.V., June, C.H., Camara, P.G., et al. (2021). Single-cell multimics dissection of basal and antigen-specific activation states of CD19-targeted CAR T cells. *J. Immunother. Cancer* 9, e002328. <https://doi.org/10.1136/jitc-2020-002328>.
18. Wilson, T.L., Kim, H., Chou, C.-H., Langfitt, D., Mettelman, R.C., Minervina, A.A., Allen, E.K., Métais, J.-Y., Pogorely, M.V., Riberdy, J.M., et al. (2022). Common Trajectories of Highly Effective CD19-Specific CAR T Cells Identified by Endogenous T-cell Receptor Lineages. *Cancer Discov.* 12, 2098–2119. <https://doi.org/10.1158/2159-8290.CD-21-1508>.
19. Sheih, A., Voillet, V., Hanafi, L.-A., DeBerg, H.A., Yajima, M., Hawkins, R., Gersuk, V., Riddell, S.R., Maloney, D.G., Wohlfahrt, M.E., et al. (2020). Clonal kinetics and single-cell transcriptional profiling of CAR-T cells in patients undergoing CD19 CAR-T immunotherapy. *Nat. Commun.* 11, 219. <https://doi.org/10.1038/s41467-019-13880-1>.
20. Jackson, Z., Hong, C., Schauner, R., Dropulic, B., Caimi, P.F., de Lima, M., Giraudo, M.F., Gupta, K., Reese, J.S., Hwang, T.H., and Wald, D.N. (2022). Sequential Single-Cell Transcriptional and Protein Marker Profiling Reveals TIGIT as a Marker of CD19 CAR-T Cell Dysfunction in Patients with Non-Hodgkin Lymphoma. *Cancer Discov.* 12, 1886–1903. <https://doi.org/10.1158/2159-8290.CD-21-1586>.
21. Haradhvala, N.J., Leick, M.B., Maurer, K., Gohil, S.H., Larson, R.C., Yao, N., Gallagher, K.M.E., Katsis, K., Frigault, M.J., Southard, J., et al. (2022). Distinct cellular dynamics associated with response to CAR-T therapy for refractory B cell lymphoma. *Nat. Med.* 28, 1848–1859. <https://doi.org/10.1038/s41591-022-01959-0>.
22. Rabilloud, T., Potier, D., Pankaew, S., Nozais, M., Loosveld, M., and Payet-Bornet, D. (2021). Single-cell profiling identifies pre-existing CD19-negative subclones in a B-ALL patient with CD19-negative relapse after CAR-T therapy. *Nat. Commun.* 12, 865. <https://doi.org/10.1038/s41467-021-21168-6>.
23. Ortiz-Maldonado, V., Rives, S., Castellà, M., Alonso-Saladrigues, A., Benítez-Ribas, D., Caballero-Baños, M., Baumann, T., Cid, J., Garcia-Rey, E., Llanos, C., et al. (2021). CART19-BE-01: A Multicenter Trial of ARI-0001 Cell Therapy in Patients with CD19+ Relapsed/Refractory Malignancies. *Mol. Ther.* 29, 636–644. <https://doi.org/10.1016/j.ymthe.2020.09.027>.
24. Damodar, S., Bhat, S., Nayak, A., Mallya, P., Ram S, B., Joshi, R., MB, D., Chougule, S., Joseph, A.R., Jakka, G., et al. (2022). Early Results from a Phase-2 Study of Varnimcabtagene Autoleucel (IMN-003A), a First-in-India Industry CD19-Directed CAR-T Cell Therapy with Fractionated Infusions for Patients with Relapsed and/or Refractory B Cell Malignancies (IMAGINE Study). *Blood* 140, 10343–10344. <https://doi.org/10.1182/blood-2022-166297>.
25. Castella, M., Caballero-Baños, M., Ortiz-Maldonado, V., González-Navarro, E.A., Suñé, G., Antóñana-Vidósola, A., Boronat, A., Marzal, B., Millán, L., Martín-Antonio, B., et al. (2020). Point-Of-Care CAR T-Cell Production (ARI-0001) Using a Closed Semi-automatic Bioreactor: Experience From an Academic Phase I Clinical Trial. *Front. Immunol.* 11, 482. <https://doi.org/10.3389/fimmu.2020.00482>.
26. Bratke, K., Kuepper, M., Bade, B., Virchow, J.C., and Luttmann, W. (2005). Differential expression of human granzymes A, B, and K in natural killer cells and during CD8+ T cell differentiation in peripheral blood. *Eur. J. Immunol.* 35, 2608–2616. <https://doi.org/10.1002/eji.200526122>.
27. Korell, F., Berger, T.R., and Maus, M.V. (2022). Understanding CAR T cell-tumor interactions: Paving the way for successful clinical outcomes. *Méd.* 3, 538–564. <https://doi.org/10.1016/j.medj.2022.05.001>.
28. Haradhvala, N.J., and Maus, M.V. (2024). Understanding Mechanisms of Response to CAR T-cell Therapy through Single-Cell Sequencing: Insights and Challenges. *Blood Cancer Discov.* 5, 86–89, OF1–OF4. <https://doi.org/10.1158/2643-3230.BCD-23-0212>.
29. Richard, A.C., Frazer, G.L., Ma, C.Y., and Griffiths, G.M. (2021). Staggered starts in the race to T cell activation. *Trends Immunol.* 42, 994–1008. <https://doi.org/10.1016/j.it.2021.09.004>.
30. Rodríguez-Otero, P., Prósper, F., Alfonso, A., Paiva, B., and San Miguel, J.F. (2020). CAR T-Cells in Multiple Myeloma Are Ready for Prime Time. *J. Clin. Med.* 9, 3577. <https://doi.org/10.3390/jcm9113577>.
31. Locke, F.L., Rossi, J.M., Neelapu, S.S., Jacobson, C.A., Miklos, D.B., Ghobadi, A., Oluwole, O.O., Reagan, P.M., Lekakis, L.J., Lin, Y., et al. (2020). Tumor burden, inflammation, and product attributes determine outcomes of axicabtagene ciloleucel in large B-cell lymphoma. *Blood Adv.* 4, 4898–4911. <https://doi.org/10.1182/bloodadvances.2020002394>.
32. Kim, M.Y., Jayasinghe, R., Devenport, J.M., Ritchey, J.K., Rettig, M.P., O'Neal, J., Staser, K.W., Kennerly, K.M., Carter, A.J., Gao, F., et al. (2022). A long-acting interleukin-7, rIL-7-hyFc, enhances CAR T cell expansion, persistence, and anti-tumor activity. *Nat. Commun.* 13, 3296. <https://doi.org/10.1038/s41467-022-30860-0>.
33. Castella, M., Boronat, A., Martín-Ibáñez, R., Rodríguez, V., Suñé, G., Caballero, M., Marzal, B., Pérez-Amill, L., Martín-Antonio, B., Castañón, J., et al. (2019). Development of a Novel Anti-CD19 Chimeric Antigen Receptor: A Paradigm for an Affordable CAR T Cell Production at Academic Institutions. *Mol. Ther. Methods Clin. Dev.* 12, 134–144. <https://doi.org/10.1016/j.omtm.2018.11.010>.
34. Wang, X., Peticone, C., Kotsopoulou, E., Göttgens, B., and Calero-Nieto, F.J. (2021). Single-cell transcriptome analysis of CAR T-cell products reveals subpopulations, stimulation, and exhaustion signatures. *Oncol Immunology* 10, 1866287. <https://doi.org/10.1080/2162402X.2020.1866287>.
35. Galli, E., Bellesi, S., Pansini, I., Di Cesare, G., Iacovelli, C., Malafronte, R., Maiolo, E., Chiusolo, P., Sica, S., Sorà, F., and Hohaus, S. (2023). The CD4/CD8 ratio of infused CD19-CAR-T is a prognostic factor for efficacy and toxicity. *Br. J. Haematol.* 203, 564–570. <https://doi.org/10.1111/bjh.19117>.
36. Shah, N.N., Highfill, S.L., Shalabi, H., Yates, B., Jin, J., Wolters, P.L., Ombrillo, A., Steinberg, S.M., Martin, S., Delbrook, C., et al. (2020). CD4/CD8 T-Cell Selection Affects Chimeric Antigen Receptor (CAR) T-Cell Potency and Toxicity: Updated Results From a Phase I Anti-CD22 CAR T-Cell Trial. *J. Clin. Oncol.* 38, 1938–1950. <https://doi.org/10.1200/JCO.19.03279>.
37. Gumber, D., and Wang, L.D. (2022). Improving CAR-T immunotherapy: Overcoming the challenges of T cell exhaustion. *EBioMedicine* 77, 103941. <https://doi.org/10.1016/j.ebiom.2022.103941>.
38. Yin, X., He, L., and Guo, Z. (2023). T-cell exhaustion in CAR-T-cell therapy and strategies to overcome it. *Immunology* 169, 400–411. <https://doi.org/10.1111/imm.13642>.
39. Bulliard, Y., Andersson, B.S., Baysal, M.A., Damiano, J., and Tsimberidou, A.M. (2023). Reprogramming T cell differentiation and exhaustion in CAR-T cell therapy. *J. Hematol. Oncol.* 16, 108. <https://doi.org/10.1186/s13045-023-01504-7>.
40. Gardner, R.A., Finney, O., Annesley, C., Brakke, H., Summers, C., Leger, K., Bleakley, M., Brown, C., Mgebroff, S., Kelly-Spratt, K.S., et al. (2017). Intent-to-treat leukemia remission by CD19 CAR T cells of defined formulation and dose in children and young adults. *Blood* 129, 3322–3331. <https://doi.org/10.1182/blood-2017-02-769208>.
41. Liao, Y.-M., Hsu, S.-H., and Chiou, S.-S. (2024). Harnessing the Transcriptional Signatures of CAR-T-Cells and Leukemia/Lymphoma Using Single-Cell Sequencing Technologies. *Int. J. Mol. Sci.* 25, 2416. <https://doi.org/10.3390/ijms25042416>.

42. Martín-Martín, L., Gutiérrez-Herrero, S., Herrero-García, M., Martín García-Sancho, A., Yeguas, A., Martín-López, A.-Á., López-Corral, L., Pérez-López, E., García-Blázquez, M., Sánchez-Guijo, F., et al. (2024). Impact of the kinetics of circulating anti-CD19 CAR-T cells and their populations on the outcome of DLBCL patients. *Blood Cancer J.* 14, 1–5. <https://doi.org/10.1038/s41408-024-01065-z>.
43. Garcillán, B., Marin, A.V.M., Jiménez-Reinoso, A., Briones, A.C., Muñoz-Ruiz, M., García-León, M.J., Gil, J., Allende, L.M., Martínez-Naves, E., Toribio, M.L., et al. (2015). $\gamma\delta$ T Lymphocytes in the Diagnosis of Human T Cell Receptor Immunodeficiencies. *Front. Immunol.* 6, 20. <https://doi.org/10.3389/fimmu.2015.00020>.
44. Hunter, S., Willcox, C.R., Davey, M.S., Kasatskaya, S.A., Jeffery, H.C., Chudakov, D.M., Oo, Y.H., and Willcox, B.E. (2018). Human liver infiltrating $\gamma\delta$ T cells are composed of clonally expanded circulating and tissue-resident populations. *J. Hepatol.* 69, 654–665. <https://doi.org/10.1016/j.jhep.2018.05.007>.
45. Mensurado, S., Blanco-Domínguez, R., and Silva-Santos, B. (2023). The emerging roles of $\gamma\delta$ T cells in cancer immunotherapy. *Nat. Rev. Clin. Oncol.* 20, 178–191. <https://doi.org/10.1038/s41571-022-00722-1>.
46. Louie, R.H.Y., Cai, C., Samir, J., Singh, M., Deveson, I.W., Ferguson, J.M., Amos, T.G., McGuire, H.M., Gowrishankar, K., Adikari, T., et al. (2023). CAR+ and CAR– T cells share a differentiation trajectory into an NK-like subset after CD19 CAR T cell infusion in patients with B cell malignancies. *Nat. Commun.* 14, 7767. <https://doi.org/10.1038/s41467-023-43656-7>.
47. Rozenbaum, M., Meir, A., Aharon, Y., Itzhaki, O., Schachter, J., Bank, I., Jacoby, E., and Besser, M.J. (2020). Gamma-Delta CAR-T Cells Show CAR-Directed and Independent Activity Against Leukemia. *Front. Immunol.* 11. <https://doi.org/10.3389/fimmu.2020.01347>.
48. Ng, J.W.K., Tan, K.W., Guo, D.Y., Lai, J.J.H., Fan, X., Poon, Z., Lim, T.H., Lim, A.S.T., Lim, T.K.H., Hwang, W.Y.K., et al. (2023). Cord blood-derived $V_{\alpha}2^{+}$ and $V_{\alpha}2^{-}$ T cells acquire differential cell state compositions upon in vitro expansion. *Sci. Adv.* 9, ead3120. <https://doi.org/10.1126/sciadv.adf3120>.
49. Hao, Y., Hao, S., Andersen-Nissen, E., Mauck, W.M., Zheng, S., Butler, A., Lee, M.J., Wilk, A.J., Darby, C., Zager, M., et al. (2021). Integrated analysis of multimodal single-cell data. *Cell* 184, 3573–3587.e29. <https://doi.org/10.1016/j.cell.2021.04.048>.
50. McGinnis, C.S., Murrow, L.M., and Gartner, Z.J. (2019). DoubletFinder: Doublet Detection in Single-Cell RNA Sequencing Data Using Artificial Nearest Neighbors. *Cell Syst.* 8, 329–337.e4. <https://doi.org/10.1016/j.cels.2019.03.003>.
51. Germain, P.L., Lun, A., Garcia Meixide, C., Macnair, W., and Robinson, M.D. (2021). Doublet identification in single-cell sequencing data using scDbtFinder. *F1000Res.* 10, 979.
52. Korsunsky, I., Millard, N., Fan, J., Slowikowski, K., Zhang, F., Wei, K., Baglaenko, Y., Brenner, M., Loh, P.R., and Raychaudhuri, S. (2019). Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat. Methods* 16, 1289–1296. <https://doi.org/10.1038/s41592-019-0619-0>.
53. Stuart, T., Butler, A., Hoffman, P., Hafemeister, C., Papalexi, E., Mauck, W.M., Hao, Y., Stoeckius, M., Smibert, P., and Satija, R. (2019). Comprehensive Integration of Single-Cell Data. *Cell* 177, 1888–1902.e21. <https://doi.org/10.1016/j.cell.2019.05.031>.
54. Aran, D., Looney, A.P., Liu, L., Wu, E., Fong, V., Hsu, A., Chak, S., Naikawadi, R.P., Wolters, P.J., Abate, A.R., et al. (2019). Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage. *Nat. Immunol.* 20, 163–172. <https://doi.org/10.1038/s41590-018-0276-y>.
55. Chu, Y., Dai, E., Li, Y., Han, G., Pei, G., Ingram, D.R., Thakkar, K., Qin, J.-J., Dang, M., Le, X., et al. (2023). Pan-cancer T cell atlas links a cellular stress response state to immunotherapy resistance. *Nat. Med.* 29, 1550–1562. <https://doi.org/10.1038/s41591-023-02371-y>.
56. Carmona, S.J., Siddiqui, I., Bilous, M., Held, W., and Gfeller, D. (2020). Deciphering the transcriptomic landscape of tumor-infiltrating CD8 lymphocytes in B16 melanoma tumors with single-cell RNA-Seq. *Oncotarget* 9, 1737369. <https://doi.org/10.1080/2162402X.2020.1737369>.
57. Mangiola, S., Schulze, A., Trussart, M., Zozaya, E., Ma, M., Gao, Z., Rubin, A., Speed, T., Shim, H., and Papenfuss, A. (2022). Robust differential composition and variability analysis for multisample cell omics. *Bioinformatics*. <https://doi.org/10.1101/2022.03.04.482758>.
58. Leek, J.T., Johnson, W.E., Parker, H.S., Jaffe, A.E., and Storey, J.D. (2012). The sva package for removing batch effects and other unwanted variation in high-throughput experiments. *Bioinformatics* 28, 882–883. <https://doi.org/10.1093/bioinformatics/bts034>.
59. Love, M.I., Huber, W., and Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* 15, 550. <https://doi.org/10.1186/s13059-014-0550-8>.
60. Andreatta, M., and Carmona, S.J. (2021). UCell: Robust and scalable single-cell gene signature scoring. *Comput. Struct. Biotechnol. J.* 19, 3796–3798. <https://doi.org/10.1016/j.csbj.2021.06.043>.
61. Subramanian, A., Tamayo, P., Mootha, V.K., Mukherjee, S., Ebert, B.L., Gillette, M.A., Paulovich, A., Pomeroy, S.L., Golub, T.R., Lander, E.S., and Mesirov, J.P. (2005). Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc. Natl. Acad. Sci. USA* 102, 15545–15550.
62. Trapnell, C., Cacchiarelli, D., Grimsby, J., Pokharel, P., Li, S., Morse, M., Lennon, N.J., Livak, K.J., Mikkelsen, T.S., and Rinn, J.L. (2014). The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells. *Nat. Biotechnol.* 32, 381–386. <https://doi.org/10.1038/nbt.2859>.
63. Kirsch, I., Vignali, M., and Robins, H. (2015). T-cell receptor profiling in cancer. *Mol. Oncol.* 9, 2063–2070. <https://doi.org/10.1016/j.molonc.2015.09.003>.
64. Subramanian, A., Tamayo, P., Mootha, V.K., Mukherjee, S., Ebert, B.L., Gillette, M.A., Paulovich, A., Pomeroy, S.L., Golub, T.R., Lander, E.S., and Mesirov, J.P. (2005). Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles. *Proc. Natl. Acad. Sci. USA* 102, 15545–15550. <https://doi.org/10.1073/pnas.0506580102>.
65. Zhang, L., Yu, X., Zheng, L., Zhang, Y., Li, Y., Fang, Q., Gao, R., Kang, B., Zhang, Q., Huang, J.Y., et al. (2018). Lineage tracking reveals dynamic relationships of T cells in colorectal cancer. *Nature* 564, 268–272. <https://doi.org/10.1038/s41586-018-0694-x>.
66. Gu, Z., Gu, L., Eils, R., Schlesner, M., and Brors, B. (2014). circlize implements and enhances circular visualization in R. *Bioinformatics* 30, 2811–2812. <https://doi.org/10.1093/bioinformatics/btu393>.
67. Wu, T.D., Madireddi, S., de Almeida, P.E., Bancheau, R., Chen, Y.-J.J., Chitre, A.S., Chiang, E.Y., Iftikhar, H., O’Gorman, W.E., Au-Yeung, A., et al. (2020). Peripheral T cell expansion predicts tumour infiltration and clinical response. *Nature* 579, 274–278. <https://doi.org/10.1038/s41586-020-2056-8>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
CD3-FITC	BD Bioscience	555332; RRID:AB_395739
CD4-PE	BD Bioscience	555347; RRID:AB_395752
CD8-PerCP Cy5.5	BD Bioscience	560662; RRID:AB_1727513
CD19-BV421	BD Bioscience	562440; RRID:AB_11153299
TCR $\gamma\delta$ -PECy7	BD Biosciences	655410; RRID:AB_2870377
CD19 protein (20–291)-FITC	ACRO Biosystems	CD9-HF2H2
CD4-BV421	BD Biosciences	562424; RRID:AB_11154417
CD8-BV510	BD Biosciences	563919; RRID:AB_2722546
LAG3-PE	BD Biosciences	565716; RRID:AB_2744328
PD1-APC	BD Biosciences	558694; RRID:AB_1645458
CTLA4- PE	BD Biosciences	553720; RRID:AB_395005
TIM-3-PE	BD Biosciences	563422; RRID:AB_2716866
Biological samples		
Manufactured infusion product	Hospital Clinic de Barcelona	N/A
Peripheral blood from CAR T treated patients	Hospital Clinic de Barcelona	N/A
Deposited data		
scRNAseq data	NCBI Gene expression omnibus (GEO)	GSE235760
Browsable analyzed data	CellXGene	https://cellxgene.cziscience.com/collections/14dc301f-d4fb-4743-a590-aa88d5f1df1a
Code scRNAseq	GitHub	Zenodo https://doi.org/10.5281/zenodo.13845259
Code scTCRseq	GitHub	Zenodo https://doi.org/10.5281/zenodo.13847747
Software and algorithms		
Cell Ranger software	10x Genomics	v3.1.0
R	https://www.r-project.org/	v4.0.5
Seurat	CRAN	v4.3.0
DoubletFinder	GitHub	v2.0.4
scDbfFinder	Bioconductor	v1.16.0
Harmony	GitHub	v1.2.0
Azimuth	https://azimuth.hubmapconsortium.org/	v0.5.0
singleR	Bioconductor	v2.4.1
sccomp	Bioconductor	v1.7.12
clusterProfiler	Bioconductor	v4.8.3
sva	Bioconductor	v3.48.0
DESeq2	Bioconductor	v1.40.2
UCell	GitHub	v2.4.0
msigdbR	Bioconductor	v7.5.1
Monocle2	Bioconductor	v2.32.0
scRepertoire	Bioconductor	v2.0.5
startrac	GitHub	v0.1.0
survival	CRAN	v3.5-8
survminer	CRAN	v0.4.9

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Patient samples and sample collection

In the discovery cohort, we analyzed T cells from five consecutive adult r/r B-ALL patients treated with the CD19CAR-T cell product varnimcabtagene autoleucel (varni-cel)^{8,23,24}. The patients provided written informed consent and this study has been approved by the Research Ethics Committee (Celm) of Hospital Clinic. HCB/2017/0001 according to institutional guidelines and the principles for the declaration of Helsinki. CAR-T cells were successfully manufactured from leukapheresis from all patients enrolled in the study. After 1:1 CD4:CD8 T cell selection, the cells were activated using anti-CD3/anti-CD28 beads and transduced with the CD19-specific CAR using a lentiviral vector. Following transduction, the cells were expanded *ex vivo* for 8 (+/−1) days, and 10⁶ CAR-T cells/kg body weight was infused into each patient in three sequential administrations (10:30:60).²⁵ All patients received cyclophosphamide (900 mg/m²) and fludarabine (90 mg/m²) lymphodepleting chemotherapy prior to CAR-T cell infusion, as per protocol.²³ Following CAR-T cell infusion, patients were monitored using FACS in both PB (weekly for the first 5 weeks post-infusion) and BM to track disease recurrence and CAR-T cell persistence, as previously described.²⁵ Mononuclear cells (MNCs) were isolated by density gradient centrifugation from the manufactured IP (pre-infusion) and from PB the week of CAR-T cell expansion peak following infusion (peak). Detailed clinical and biological information of patients from the discovery cohort is summarized in Table S1. Additional 65 r/r B-ALL patients treated with varni-cel in the context of the original phase I trial, compassionate use, or hospital exemption,^{8,23,24} as well as 58 r/r DLBCL patients treated with axi-cel (*n* = 33) and tisa-cel (*n* = 25) were included in the expanded validation cohorts (Table S2). This study was approved by the Institutional Review Boards of the Hospital Clinic of Barcelona (r/r B-ALL) and University Hospital of Salamanca (r/r DLBCL).

METHOD DETAILS

FACS sorting

MNCs from either IP or peak were immunophenotyped using FACS and the monoclonal antibodies CD3-FITC (BD Biosciences, San Jose, CA), CD4-PE (BD Biosciences), and CD8-PerCP Cy5.5 (BD Biosciences). The cell surface expression of the CD19CAR was traced by using AffiniPure F(ab')₂ fragment goat anti-mouse IgG (H + L)-APC (Jackson ImmunoResearch, Westgrove, PA). CAR^{pos} and CAR^{neg} CD3⁺ T cells were FACS-sorted at high purity (>95%) using a FACS Aria flow cytometer (BD Biosciences).

Library preparation and scRNA-seq and sc-αβTCR-seq

CAR^{pos} and CAR^{neg} CD3⁺ T-cells were centrifuged for 5 min at 500×*g* at 4°C to a concentration of 700–1200 cells/μL. Cell concentration and viability were verified using the TC20 Automated Cell Counter (Bio-Rad Laboratories, Hercules, CA) after staining the cells with Trypan blue. Cells were partitioned into Gel Bead-in-Emulsions (GEMs) with a target cell recovery of 5000 total cells per sample using the Chromium Controller system (10× Genomics, Pleasanton, CA). Gene expression (GEX) and TCR-enriched sequencing libraries were prepared using the Chromium Single Cell 5' Library and Gel Bead Kit v1.0 (10× Genomics, #1000006) and the Chromium Single Cell V(D)J Enrichment human T cell kit (10× Genomics, #1000005), respectively. Briefly, after GEM-RT clean up, cDNA was amplified for 14 cycles and quantified with an Agilent Bioanalyzer High Sensitivity Chip (Agilent Technologies, Santa Clara, CA). GEX libraries were prepared using 1–25 ng of cDNA and indexed with 16 cycles of PCR using the PN-220103 Chromium i7 Sample Index Plate. For the preparation of αβTCR-enriched libraries, 11 cycles of TCR target enrichment PCR 1 and 2 were performed and the resulting cDNA was quantified on an Agilent Bioanalyzer High Sensitivity chip (Agilent Technologies). TCR libraries were prepared and indexed with 9 cycles of amplification using the PN-220103 Chromium i7 Sample Index Plate. The size distribution and concentration of the libraries were verified on an Agilent Bioanalyzer High Sensitivity chip (Agilent Technologies). Finally, sequencing of GEX and TCR libraries was carried out on an Illumina NovaSeq 6000 sequencer to obtain approximately 20,000 reads per cell and 5000 reads per cell, respectively.

Processing of scRNA-seq and scTCR-seq

Cell Ranger software (10× Genomics; v3.1.0) was used to demultiplex the BCL files into FASTQ files using the cellranger mkfastq pipeline. The different FASTQ files were aligned to the reference genome (version hg38) with the addition of the CAR construct sequence using the cellranger count pipeline. The Cell Ranger Single-Cell Software Suite (v3.1.0) with VDJ option was used to map TCR-enriched libraries to the GRCh38 VDJ pre-built reference provided by 10x Genomics. Cell Ranger performed default filtering for quality control, and produced barcodes.tsv, genes.tsv, and matrix.mtx files for each sample. Clonotype data were included into the metadata of each sample. A total of 38,190 T cells and 31,863 productive TCR sequences were obtained (Tables S3 and S4).

scRNA-seq data analysis

The following downstream analyses were performed with Seurat library (v4.3.0)⁴⁹ for R 4.0.5. Prior to data integration, each sample was analyzed individually, and quality control and filtering steps were performed to remove potential apoptotic or damaged cells, as well as doublets, following well-established protocols. The filtered-out cells met any of the following criteria: <200 genes detected, >20% of genes expressed were mitochondrial genes or log₁₀(UMI counts) > 3. Doublets were identified using both DoubletFinder

(v2.0.4)⁵⁰ and scDbfFinder(v1.16.0)⁵¹ cells identified as doublets by both softwares were excluded from the analysis. In addition, a very limited number (<1%) of myeloid cells expressing CD14 or LYZ were detected and excluded from the analyses. A few contaminating CAR^{pos} T cells found within the FACS-sorted CAR^{neg} T cell fraction were also excluded. A total of 37,100 T cells passed the quality controls with an average of 7,420 cells per patient. 18,978 and 18,122 of the sequenced T cells were CAR^{pos} and CAR^{neg}, respectively. 18,492 and 18,608 of the sequenced T cells were from IP and peak, respectively (Tables S2 and S3). After filtering, the LogNormalize method was used to normalize the expression data, and the 3000 most variable genes were obtained with FindVariableFeatures. Harmony (v1.2.0)⁵² package was used for data integration, with timepoint and sample information as covariates. The integrated dataset was scaled using the ScaleData function to normalize the mean expression across cells to 0. The first 30 principal components were used for linear dimensionality reduction following a PCA exploration. Cell clusters were then identified using the FindNeighbors and FindClusters functions, with a resolution of 0.7, and visualized with the UMAP algorithm via RunUMAP, employing the newly calculated Harmony reduction.

CD4 and CD8 projection

The classification of CD4 and CD8 T cells based on direct gene expression analysis can be challenging due to the sparsity of scRNA-seq data. To address this issue, we utilized projection-based classification methods by leveraging previously published reference datasets and KNN-smooth imputation. Specifically, we used: (i) Azimuth,⁴⁹ a software with a PBMC dataset⁵³ comprising 11,769 cells that were sequenced by scRNA-seq and CITE-seq; (ii) singleR software⁵⁴ to transfer labels from Pan cancer T cell atlas,⁵⁵ which contained 308,048 sequenced T cells; and (iii) KNN-smoothed method to impute the expression of CD4 and CD8A. To ensure reliable classification, we required consensus among Azimuth, singleR, and KNN-smooth for a cell to be classified as CD4 or CD8. Cells that were not classified or showed disagreement between the methods were excluded from categorization.

Cluster analysis, cell type classification and signature analysis

The first 30 principal components, with a resolution of 0.7, were used to identify transcriptionally distinct cellular clusters. For annotation, the most variable markers of each cluster were computed using the FindAllMarkers function, configured to return only positive markers with a minimum of 0.25% and a logFC threshold of 0.5. Cluster annotations were performed based on DEGs and were manually identified using canonical markers and gene sets from the literature (Table S5).^{55,56} Composition analysis was performed to visualize differences in cluster abundances between conditions using the sccomp package (v1.0.0).⁵⁷ Differences in GEX between conditions was also assessed, through a pseudobulk analysis which involved aggregating the gene expression by each sample condition using the AggregateExpression function from Seurat. Patient batch effects were corrected using ComBat_seq from the sva package (v 3.48),⁵⁸ and differential expression analysis was then performed with DESeq2 (v 1.40.2)⁵⁹ following the standard procedure. The expression of different transcriptomic signatures was assessed to characterize the differences among patients, timepoints, conditions and cell types. The UCell suite⁶⁰ was used to assign a score to each gene set (Table S4). Gene sets were collected either from misigDB⁶¹ or from previous literature.

Trajectory analysis

Trajectory analyses were performed using the Monocle2 package (v2.24.1)⁶² to order cells in pseudotime, serving as a measure of progress through biological processes based on transcriptional similarities. The analyses focused exclusively on CAR^{pos} T cells and began by creating a “CellDataSet” object using a negative binomial model. The Monocle algorithm was then applied to the 3,000 most highly variable genes selected with Seurat after DDRTree dimension reduction and cell ordering. To visualize the ordered cells in the trajectory, the plot_cell_trajectory function was used to plot the minimum spanning tree on the cells.

Sc- $\alpha\beta$ TCR repertoire analysis

The sc- $\alpha\beta$ TCR repertoire sequencing data were analyzed using the scRepertoire package (v1.7.1). Filtered Cell Ranger annotations were imported, and TCR $\alpha\beta$ chains were combined for all T cells based on their cell barcodes using the combineTCR function. Additionally, the combineExpression function was employed to integrate the combined $\alpha\beta$ TCR data with the scRNA-seq data. Clonotypes were categorized by the absolute number (x) of T cells expressing individual clonotype sequences (clonotype) as single (x = 1 clonotype), small (1 < x ≤ 5 clonotypes), medium (5 < x ≤ 20 clonotypes), large (20 < x ≤ 100 clonotypes) and hyperexpanded (x > 100 clonotypes), thus reflecting the actual clonal expansion for each sample. Occupied TCR repertoire space was calculated for each CAR^{pos} T cell subtype by counting the number of cells in each of the mentioned expansion categories. To assess the quality and quantity of the TCR repertoire in each CAR^{pos} sample, diversity was measured using clonalDiversity function and Shannon index metric.⁶³

To evaluate the expansion and ongoing proliferation of the CAR^{pos} T cell subtypes, a proliferation and expansion index was calculated. The hallmark genes involved in the G2/M checkpoint (GSEA, misigdb⁶⁴) were used to calculate the proliferation index using the AddModuleScore function from Seurat.⁴⁹ The startrac software was also employed to calculate the expansion index.⁶⁵ The Morisita index was estimated using the overlap coefficient method to measure the similarity between clonotypes across different samples. The circlize package was employed to visualize the interconnections of shared expanded clonotypes among different T cell clusters.⁶⁶ Additionally, trend lines were drawn on scatterplots to illustrate the distribution of expanded clonotypes shared between the IP and Peak phase.⁶⁷

FACS cytometry in validation cohorts

To validate the clinical impact of CAR^{pos} $\gamma\delta$ T cells, IP samples from 58 DLBCL patients, treated with either axi-cel ($n = 33$) or tisa-cel ($n = 25$), and one-year post-treatment PB samples from 18 B-ALL patients treated with varni-cel were analyzed. Samples were stained using the EuroFlow Lymphocyte Screening Tube (LST), along with the immune monitoring T CD4⁺ (TCD4) and T/NK-cell cytotoxic (Cytox) tubes.^{5,42} The PEcy7-conjugated TCR $\gamma\delta$ antibody (clone 11F2, BD Biosciences) was used for $\gamma\delta$ T cell detection. Additionally, the CD19 protein (20–291)-FITC reagent from ACROBiosystems (Bethesda, MD) was included in the staining process. Sample staining was performed according to the EuroFlow standard procedures available at www.euroflow.org. On the other hand, IP samples from B-ALL patients ($n = 47$) treated with varni-cel were stained with fluorochrome-conjugated monoclonal antibodies against CD4, CD8, LAG3, PD-1, CTLA4, and TIM3 (all from BD Biosciences) to validate the clinical impact of the CD4:CD8 ratio and exhaustion signatures. Data acquisition and analysis were performed in a FACS Canto-II flow cytometer equipped with FACSDiva software.

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were performed using R (v4.0.5). Statistical significance was calculated using either a two-sided Chi-squared test or Fisher's exact test, as appropriate. To assess the association between the CD4:CD8 ratio and EFS, we employed a Cox proportional hazards model. Kaplan-Meier survival curves were generated using the survival package (v3.5-5) in R (v.4.3.0). Where not indicated in the text, differences were considered not statistically significant.